Project Report

Course Code: CSE475

Course Title: Machine Learning

Section: 06

Semester: Summer 2026

Project Title: Explainable Machine Learning model for Skin-Cancer Detection

Submitted by (Group-2)

Student Name	Student ID
Md. Rahat Khan	2021-1-60-090
Jannatul Alam Shifa	2022-2-60-147
Md Ahsiul Karim	2022-3-60-074
Fahim Hossain	2022-2-60-142

Submitted To

Dr. Raihan Ul Islam

Associate Professor, Department of Computer Science and Engineering,

East West University

Date of Submission: 21, August 2026

Abstract

Skin cancer is a major public health concern in which early and reliable diagnosis can significantly improve treatment outcomes. This study investigates an explainable deep-learning framework for multi-class skin lesion classification using the HAM10000 dataset, comprising 10,015 dermoscopic images from seven diagnostic categories. Four baseline architectures—ResNet50, EfficientNet-B2, Vision Transformer (ViT-B/16), and Swin-Tiny—were evaluated and compared with a graph neural network (GNN)-based approach. To reduce data leakage, image partitions were constructed at the lesion level so that images belonging to the same lesion were not shared across training and evaluation sets. The CNN- and Transformer-based models used transfer learning with class-imbalance handling, data augmentation, validation-based model selection, and model-specific fine-tuning strategies. ViT-B/16 and Swin-Tiny additionally employed lesion-aware five-fold cross-validation and probability ensembling. For the GNN pipeline, spatial features extracted from ResNet50 were converted into a graph in which local image regions were represented as nodes and processed using a two-layer GATv2 architecture selected through sequential ablation experiments. Model performance was assessed using accuracy, precision, recall, F1-score, confusion matrices, and ROC analysis, with particular emphasis on Macro-F1 because of the substantial class imbalance in HAM10000. Among the models evaluated on the common 1,506-image lesion-level test set, the ViT-B/16 ensemble achieved the strongest performance with 86.52% accuracy and a Macro-F1 score of 0.7777, followed by Swin-Tiny with 85.46% accuracy and 0.7289 Macro-F1. The final GNN achieved 75.43% accuracy, 0.5453 Macro-F1, and a Macro ROC-AUC of 0.8806. Although the GNN did not outperform the Transformer baselines, its ablation study demonstrated measurable improvement over the initial GNN configuration and provided additional interpretability through SHAP and LIME. Overall, the findings highlight the strong predictive capability of Transformer-based models while demonstrating the value of leakage-aware evaluation, class-sensitive metrics, and explainable analysis for trustworthy automated skin lesion classification.

1. Introduction

Skin cancer is one of the most common diseases worldwide. Skin cancer caused by the abnormal and uncontrolled growth of skin cells is often associated with factors such as excessive ultraviolet radiation exposure. Early detection is very essential because delay can allow cancer to progress to more advanced stages and significantly affect treatment outcomes.

Conventional skin cancer diagnosis primarily relies on clinical examination and the expertise of dermatologists. However, distinguishing malignant lesions such as melanoma, basal cell carcinoma (BCC), and squamous cell carcinoma (SCC) from benign skin lesions can be challenging because many of these conditions share similar visual characteristics [1].

The increasing use of computerized medical image analysis and deep learning has created opportunities for automated skin cancer classification. Recent studies have increasingly adopted CNNs and transfer-learning architectures such as ResNet, VGG, DenseNet, and EfficientNet models. For example, Manole et al. developed an EfficientNetB3-based transfer-learning model using 8,222 skin-lesion images and reported 95.4% accuracy for four disease categories and 88.8% accuracy when six categories were considered [2]. Despite their strong performance, deep learning models often function as "black boxes," making it difficult to understand the reasoning behind their predictions. To address this limitation, Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM, Grad-CAM++, and Score-CAM can generate visual heatmaps showing the regions that contribute to a model's prediction [1].

Although deep learning shows promising results in skin cancer classification, challenges such as dataset diversity, class imbalance, generalization, computational efficiency, and interpretability remain. Therefore, our project Explainable machine learning model for skin cancer detection aims to combine effective deep learning architectures with appropriate preprocessing, data augmentation, robust evaluation, and XAI techniques. By integrating these components, the proposed framework seeks not only to achieve reliable classification performance but also to provide interpretable predictions that can support greater transparency and confidence in automated skin cancer detection.

1.1 Problem Statement

In this project, we are trying to build an accurate, low-cost, and understandable deep learning system for detecting and classifying skin cancer from images. The system should be able to look at a skin image, decide which disease it most likely shows, and explain which part of the image led to that decision. Building a system like this is not simple: it involves choosing the right dataset, designing a model that generalizes well, and making sure the model's decisions can be trusted by an actual doctor rather than treated as a black box.

2. Literature Review

Early research in automated skin disease detection mainly focused on image processing and traditional machine learning approaches. This system is very popular for several sequential stages-like image preprocessing, segmentation, feature extraction, and classification. Features working as color, texture, lesion boundaries, and shape were manually extracted and subsequently provided to classifiers such as support vector machines (SVMs) or artificial neural networks. A 2019 study by Alenezi followed this general approach by preprocessing skin images, extracting features using a pretrained CNN, and classifying those features using an SVM classifier. The study reported 100% detection accuracy for three disease categories [1].

There are two types of skin cancer, the most common cancer is NMSC, which includes mainly basal cell carcinoma (BCC) and also squamous cell carcinoma (SCC). BCC rarely metastasizes but is associated with significant morbidity in those cases that are left untreated or treated with delay. SCC, although more aggressive than BCC, usually responds well to treatment if detected and treated on time. Increased incidence of NMSCs has been associated with higher ambient UV, related to ozone depletion, and increased public awareness and screening activities [2, 8]

Melanoma is a relatively rare but highly metastatic form of skin cancer. Early detection is crucial for effective management and improving patient survival. The five-year survival rate for patients with early-stage melanoma is over 90%, but this figure decreases significantly with the higher stage at diagnosis [2, 8-9].

CNN-based approaches subsequently allowed models to learn image features automatically and relied heavily on handcrafted features. CNNs became particularly effective for medical image classification because they can learn hierarchical representations of color, texture, shape, and lesion structure. The EfficientSkinDis paper also highlights the growing importance of CNN architectures in medical image analysis [3]. The study is particularly important for addressing a broader multi-class problem detection. Its comparative results show that EfficientSkinDis achieved 87.15% Top-1 accuracy, compared with 84.70% for an EfficientNetV2 model trained on four diseases and 68.97% Top-1 accuracy reported for DenseNet on 23 skin diseases [3].

Deep CNNs have subsequently become more prominent because they can learn image representations automatically rather than relying exclusively on manually engineered features. CNN architectures consist of convolutional, pooling, activation, and fully connected layers that progressively learn increasingly complex visual patterns. The development of pretrained CNN architectures has also made transfer learning an effective approach for medical image classification, especially when the target dataset is relatively small [2], [4]. The proposed project extends this direction by considering EfficientNet-B4, ResNet-50, and DenseNet-121 as baseline architectures [4].

Recent research has expanded beyond conventional CNNs toward Vision Transformers (ViTs) and Swin Transformers. Dagnaw et al. compared two pretrained ViTs, three Swin Transformer models, and five pretrained CNN architectures, including VGG19, ResNet18, ResNet50, MobileNetV2, and DenseNet201. Their study addressed skin cancer classification using freely available datasets and employed SMOTE to handle class imbalance. The results showed that ViT-base and ViT-large achieved an accuracy of 88.6%, while ResNet50 produced the strongest achieving 88.8% accuracy [5].

Another recent direction combines GNNs with Capsule Networks. Santoso et al. proposed an approach that integrates a Tiny Pyramid Vision Graph Neural Network with a Capsule Network. The motivation was to improve the representation of complex spatial and semantic features that can be difficult for conventional CNNs to capture. The Capsule Network is intended to preserve spatial hierarchies, while GNNs model relationships between image regions. The proposed model was evaluated on the MNIST:HAM10000 dataset. After 75 epochs, the reported results reached 89.23% and 95.52% accuracy under the study's evaluated settings, with comparisons against models including GoogLeNet, InceptionV3, MobileNetV3, EfficientNet-B7, ResNet18, ResNet34, ViT-Base, and IRv2-SA [7].

GNNs provide a mechanism for representing an image as a graph and modeling relationships between different regions. This is particularly relevant to skin lesions because diagnostic information may depend not only on individual features but also on relationships between lesion regions. In this architecture, ViT extracts global contextual information, while GNN models spatial relationships among lesion regions. Region-Adaptive Attention then focuses the model on diagnostically important areas. The study also integrates SHAP and Grad-CAM to improve interpretability and evaluates the framework using HAM10000, ISIC, and PH2 datasets [6].

The study used ISIC 2020 with 33,126 images, HAM10000 with 10,015 images, and PH2 with 200 images. It also addressed class imbalance through augmentation, class-balanced sampling, and weighted cross-entropy loss [6].

Bangladesh-specific clinical image dataset developed to address the shortage of region-specific datasets for automated skin-disease classification. The dataset contains 1,710 clinical images from 416 patients, collected from two medical colleges in Bangladesh using smartphone cameras during dermatological consultations under natural lighting conditions. It covers six skin diseases: Atopic Dermatitis, Contact Dermatitis, Eczema, Scabies, Seborrheic Dermatitis, and Tinea Corporis [10]. The dataset also provides clinical and demographic metadata, including patient age, sex, lesion location, and diagnosis. To increase data diversity, the authors generated 11,970 augmented images using transformations such as rotation, flipping, scaling, translation, brightness and contrast adjustment, and shearing. However, the original dataset is considerably imbalanced, with substantially fewer images for Atopic Dermatitis and Seborrheic Dermatitis than for Contact Dermatitis and Eczema. The authors also evaluated EfficientNet-B0, ResNet-50, and VGG16, achieving accuracies of 86.93%, 86.36%, and 86.93%, respectively [10].

Despite the progress in deep-learning-based skin-disease classification, several research gaps remain. Many existing studies rely on international datasets such as HAM10000 and ISIC, which may not fully represent the clinical conditions and patient populations of Bangladesh. Although SkinDisNet provides 1,710 clinical smartphone images from Bangladesh, its relatively small size and class imbalance remain challenges for model development [10]. Previous studies have also shown that classification performance can decrease when underrepresented disease classes are included [2]. Furthermore, recent CNN-, Vision Transformer-, and GNN-based models have achieved promising results, but their generalization to heterogeneous real-world clinical images remains insufficiently investigated [5], [6]. Although Grad-CAM and other XAI methods can visualize important image regions, their clinical relevance is often not quantitatively validated against expert annotations [5]. Therefore, there is a need for a Bangladesh-focused system that combines robust CNN models, appropriate handling of class imbalance, feature fusion, and quantitative validation of Grad-CAM explanations using expert-annotated lesion regions [4].

3. Methodology

This section reconstructs the complete experimental methodology directly from the four baseline notebooks (ResNet50, EfficientNet-B2, ViT-B/16 and Swin-Tiny) and the Task-3 GNN notebook. The notebooks all use the seven-class HAM10000 dataset, but they do not all use the same test partition. Consequently, the methodology below records the exact split protocol for each model and keeps direct same-test comparisons separate from different-split reference comparisons.

3.1 Dataset, Classes, and Leakage Control

HAM10000 contains 10,015 dermatoscopic images grouped by lesion identifier (lesion_id). The seven encoded diagnosis classes are akiec, bcc, bkl, df, mel, nv and vasc. Because multiple images can correspond to the same lesion, all notebooks treat lesion_id as the grouping unit when constructing train, validation, test, or cross-validation partitions. This prevents images of the same physical lesion from appearing in both training and evaluation data.

Class	Diagnosis	Images
akiec	Actinic keratoses / intraepithelial carcinoma	327
bcc	Basal cell carcinoma	514
bkl	Benign keratosis-like lesions	1,099
df	Dermatofibroma	115
mel	Melanoma	1,113
nv	Melanocytic nevi	6,705
vasc	Vascular lesions	142

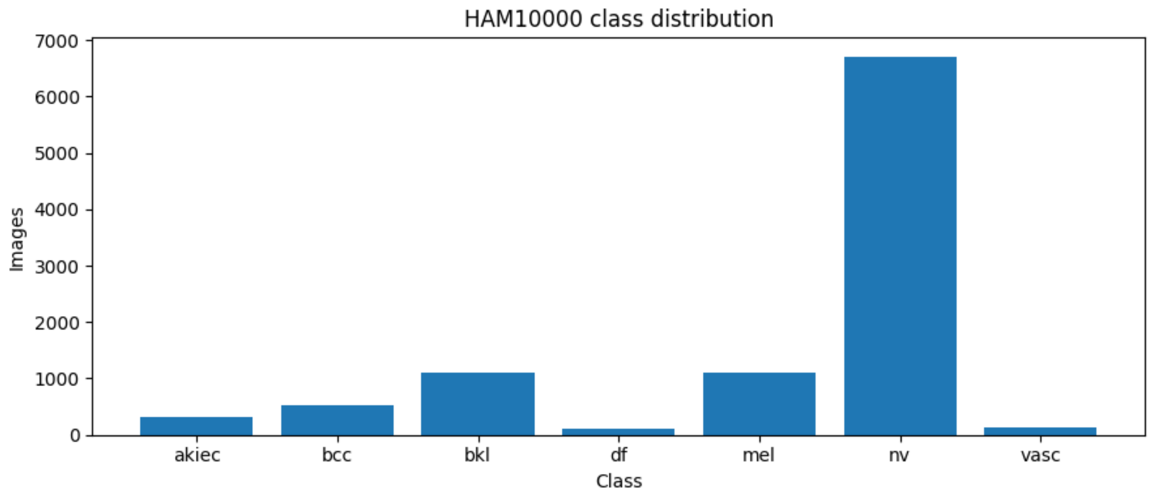

Figure 1. HAM10000 class distribution extracted from the executed GNN notebook.

3.2 Experimental Split Protocols and Comparability

ResNet50 and EfficientNet-B2 use the same seed-42 lesion-level holdout: 7,002 training images, 1,519 validation images, and 1,494 test images. ViT-B/16 and Swin-Tiny first search 200 lesion-level 15% final-test candidates and select seed 171, producing 8,509 development images and an untouched 1,506-image final test set. Five lesion-safe StratifiedGroupKFold splits are then created inside the development set. The GNN reproduces the same seed-171 1,506-image final test set used by ViT and Swin, but uses a fixed train/validation holdout (7,007/1,502 images) rather than five-fold GNN cross-validation; its selected validation split seed is 122.

Model	Train/Development	Validation/CV	Final test	Final-test seed	Ensemble
ResNet50	7,002 train	1,519 validation	1,494	42	No
EfficientNet-B2	7,002 train	1,519 validation	1,494	42	No
ViT-B/16	8,509 development	5-fold StratifiedGroupKFold	1,506	171	Mean probability of 5 fold models
Swin-Tiny	8,509 development	5-fold StratifiedGroupKFold	1,506	171	Mean probability of 5 fold models
Final GNN	7,007 train	1,502 validation	1,506	171	No

Fair-comparison rule used in this report: ViT-B/16, Swin-Tiny and the Final GNN are directly comparable on the identical seed-171 final test set. ResNet50 and EfficientNet-B2 are directly comparable with each other on their shared seed-42 test set, but their raw values should not be interpreted as strict same-test comparisons against the seed-171 models.

3.3 Common Evaluation Measures

The notebooks evaluate multi-class classification with overall accuracy, macro precision, macro recall, macro F1, weighted F1, class-wise precision, recall and F1, and confusion matrices. ViT, Swin and GNN additionally provide usable one-vs-rest ROC curves and class-wise AUC values. Macro F1 is particularly important for model selection because HAM10000 is highly imbalanced toward the nv class. For the GNN, balanced accuracy, lesion-level bootstrap confidence intervals, and ablation-stage validation Macro F1 are also reported.

3.4 Baseline Model 1: ResNet50

3.4.1 Architecture and training protocol

The ResNet50 baseline is implemented with timm using ImageNet-pretrained weights and a seven-output linear classifier (2048 to 7). The model contains 23,522,375 parameters. Inputs are resized/cropped to 224 x 224 pixels. Training augmentation uses RandomResizedCrop with scale 0.80-1.00, horizontal and vertical flipping (p=0.5), rotation up to 20 degrees, and ColorJitter with brightness/contrast/saturation 0.20 and hue 0.05. Validation and test images are deterministically resized to 224 x 224. All images are normalized with ImageNet mean and standard deviation.

Class imbalance is handled with square-root inverse-frequency class weights computed from the training split. Weighted cross-entropy with label smoothing 0.10 is used. Training runs for 50 epochs with batch size 16 and mixed precision on CUDA. During the first 5 epochs only the classifier head is trainable using AdamW with learning rate 1e-3 and weight decay 1e-4. From epoch 6 onward the full backbone is unfrozen, with backbone learning rate 1e-5 and classifier learning rate 1e-4. CosineAnnealingLR is used in both phases. The checkpoint with the highest validation Macro F1 is retained; the notebook does not use early stopping and completes the 50-epoch budget.

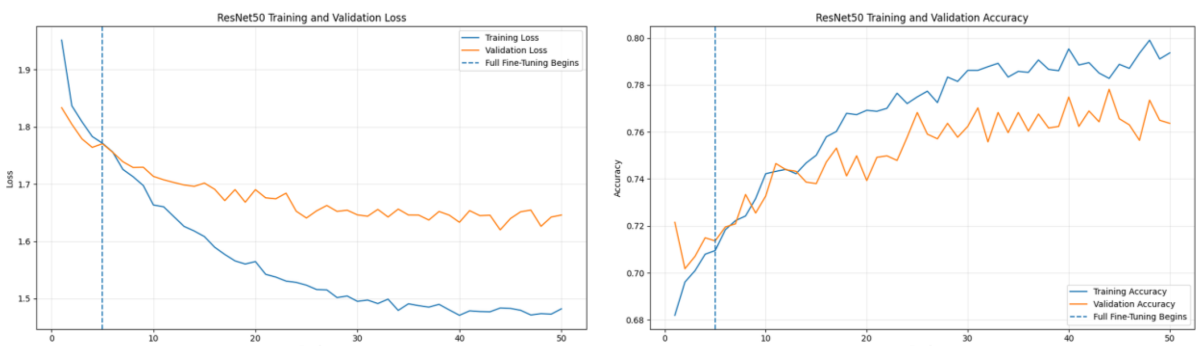

Figure 2. ResNet50 training/validation loss and accuracy curves extracted from the notebook.

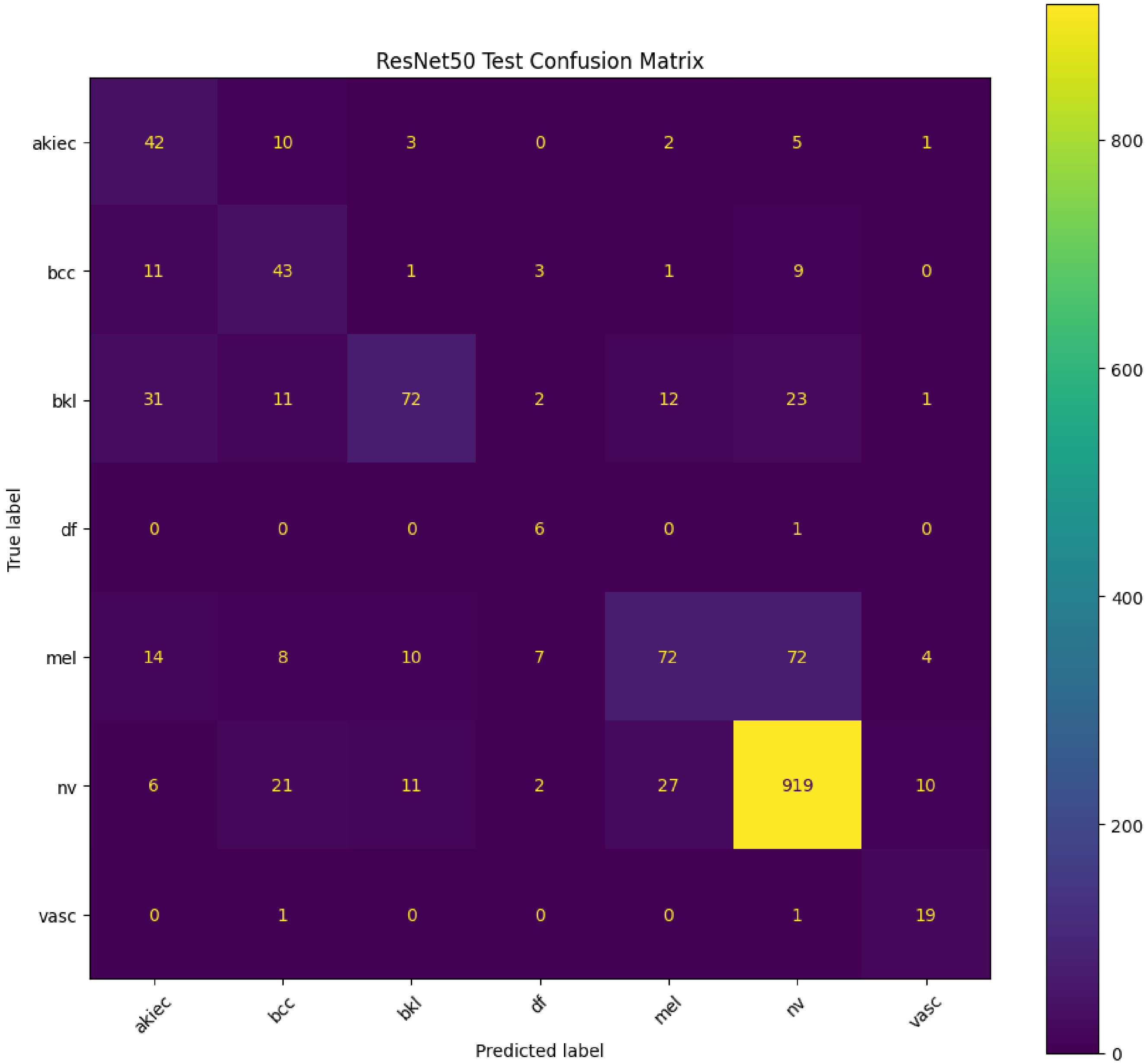

Figure 3. ResNet50 raw test confusion matrix extracted from the notebook.

3.4.2 ResNet50 evaluation output

Metric	Value
Test protocol	Seed-42 lesion-level 70/15/15 holdout
Test images	1,494
Test loss	1.6306
Overall accuracy	0.7851 (78.51%)
Macro precision	0.5672
Macro recall	0.6918
Macro F1	0.5887
Weighted F1	0.7805
Macro ROC-AUC	Not available

Class	Support	Class-wise accuracy	Precision	Recall	F1-score
akiec	63	0.6667	0.4038	0.6667	0.5030
bcc	68	0.6324	0.4574	0.6324	0.5309
bkl	152	0.4737	0.7423	0.4737	0.5783
df	7	0.8571	0.3000	0.8571	0.4444
mel	187	0.3850	0.6316	0.3850	0.4784
nv	996	0.9227	0.8922	0.9227	0.9072
vasc	21	0.9048	0.5429	0.9048	0.6786

Class-wise accuracy is defined here as the fraction of correctly classified samples within each true class (TP/support); therefore, it is numerically identical to recall for that class. This definition matches the per-class accuracy implementation used in the GNN notebook and can be read directly from the normalized confusion matrices for the image models.

The ResNet50 notebook did not generate a valid ROC curve. Its final multiclass ROC-AUC calculation raised a ValueError because the stored scores did not satisfy the probability requirement, and the aggregate AUC was recorded as None. No ROC image is present in the notebook output, so no ROC figure is invented in this report.

3.5 Baseline Model 2: EfficientNet-B2

3.5.1 Architecture and training protocol

The EfficientNet-B2 baseline uses the ImageNet-pretrained timm efficientnet_b2 architecture with a seven-output linear classifier (1408 to 7). It contains 7,710,857 parameters and uses 260 x 260 inputs. The train/validation/test split is identical to ResNet50. Training augmentation is also matched to the ResNet50 baseline: RandomResizedCrop (0.80-1.00), horizontal/vertical flips, 20-degree rotation, ColorJitter (0.20/0.20/0.20/0.05), tensor conversion, and ImageNet normalization; validation and test images are deterministically resized to 260 x 260.

The same square-root class weighting and weighted cross-entropy with 0.10 label smoothing are used. The first 5 epochs train only the EfficientNet classifier with AdamW at 1e-3; the entire network is then fine-tuned to epoch 50 with backbone learning rate 1e-5, head learning rate 1e-4, weight decay

1e-4, cosine scheduling, batch size 16, and mixed precision. The best checkpoint is selected by validation Macro F1. The notebook reports its best validation Macro F1 at epoch 50.

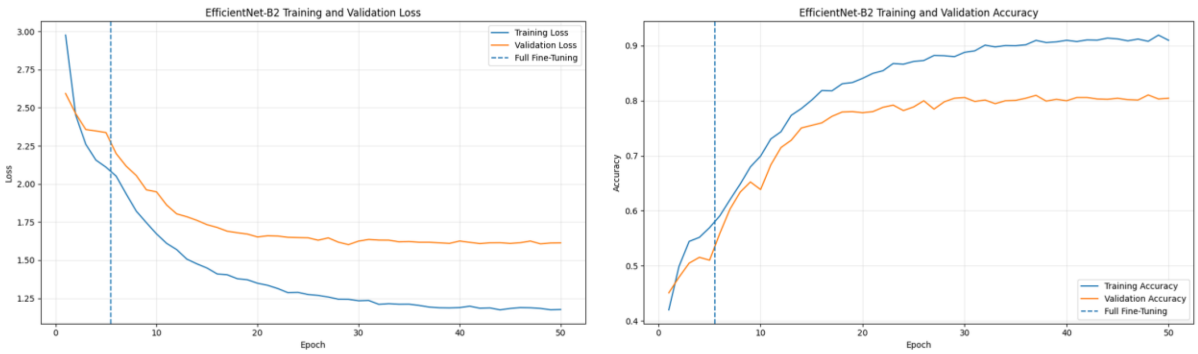

Figure 4. EfficientNet-B2 training/validation loss and accuracy curves extracted from the notebook.

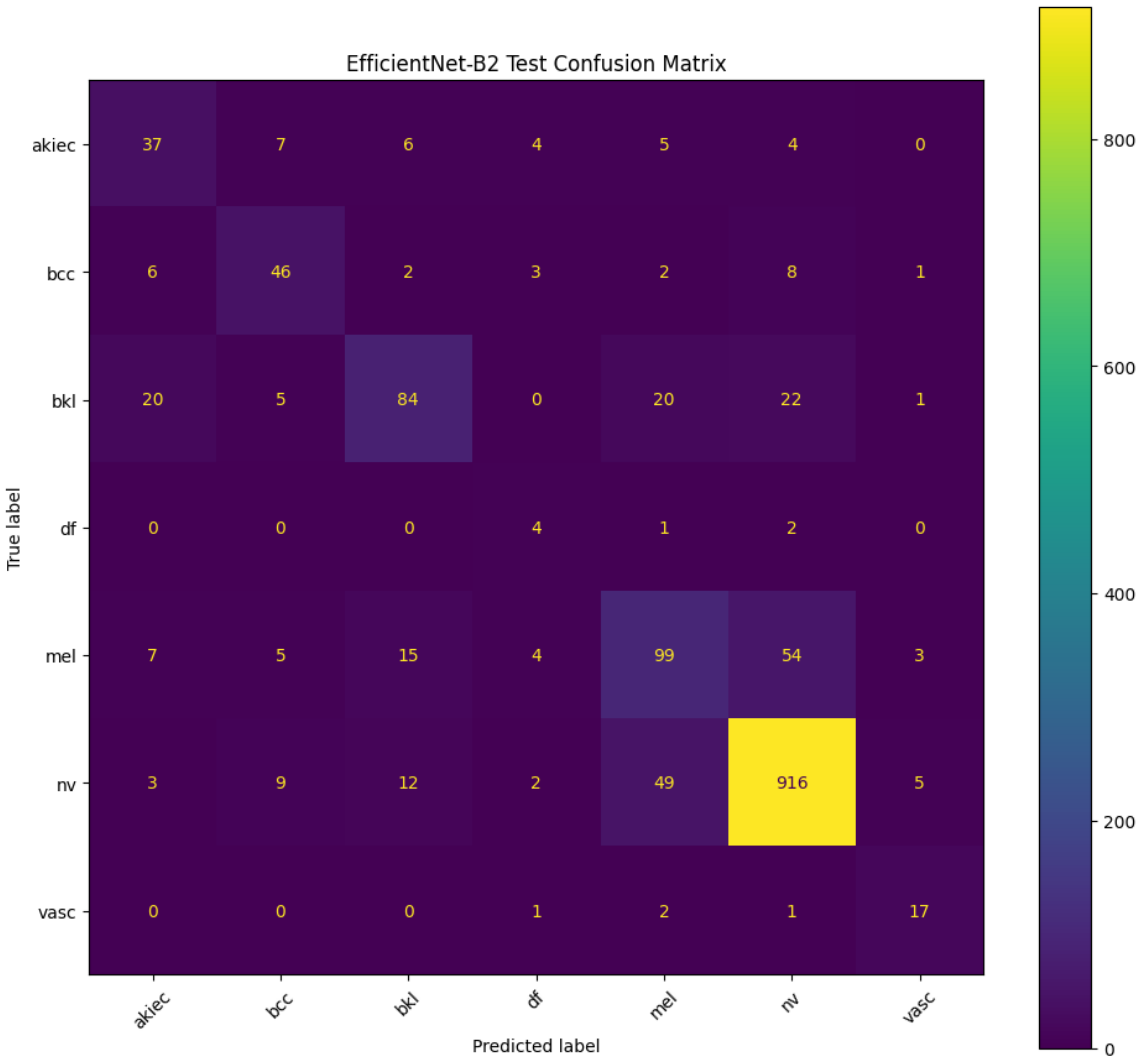

Figure 5. EfficientNet-B2 raw test confusion matrix extracted from the notebook.

3.5.2 EfficientNet-B2 evaluation output

Metric	Value
Test protocol	Seed-42 lesion-level 70/15/15 holdout
Test images	1,494
Test loss	1.6089
Overall accuracy	0.8052 (80.52%)
Macro precision	0.5956
Macro recall	0.6638
Macro F1	0.6152
Weighted F1	0.8050
Macro ROC-AUC	Not available

Class	Support	Class-wise accuracy	Precision	Recall	F1-score
akiec	63	0.5873	0.5068	0.5873	0.5441
bcc	68	0.6765	0.6389	0.6765	0.6571
bkl	152	0.5526	0.7059	0.5526	0.6199
df	7	0.5714	0.2222	0.5714	0.3200
mel	187	0.5294	0.5562	0.5294	0.5425
nv	996	0.9197	0.9096	0.9197	0.9146
vasc	21	0.8095	0.6296	0.8095	0.7083

Class-wise accuracy is defined here as the fraction of correctly classified samples within each true class (TP/support); therefore, it is numerically identical to recall for that class. This definition matches the per-class accuracy implementation used in the GNN notebook and can be read directly from the normalized confusion matrices for the image models.

As with ResNet50, the EfficientNet-B2 notebook did not store a valid final ROC curve. The multiclass ROC-AUC calculation raised the same probability-format ValueError and the aggregate AUC was saved as None; therefore no ROC image can be faithfully extracted from this notebook.

3.6 Baseline Model 3: ViT-B/16

3.6.1 Architecture, five-fold training, and ensemble

The ViT baseline uses timm model vit_base_patch16_224.augreg_in21k_ft_in1k with a 768-to-7 classifier, 85,804,039 total parameters, model dropout 0.10, and drop-path rate 0.10. Inputs are 224 x 224. The untouched final test split is selected from 200 lesion-level candidates; seed 171 is chosen because it closely matches the full class distribution while preserving all seven classes. The remaining 8,509 images are partitioned into five StratifiedGroupKFold folds with zero lesion overlap.

Fold training uses a WeightedRandomSampler with inverse class-frequency sample weights. Training augmentation uses RandomResizedCrop (scale 0.75-1.00, aspect ratio 0.90-1.10), horizontal and vertical flips, 20-degree rotation, moderate ColorJitter, Gaussian blur with probability 0.10, and ImageNet normalization. Validation and final test preprocessing use Resize(256), CenterCrop(224), and ImageNet normalization. Each fold has a maximum of 50 epochs and batch size 16. The first 5 epochs train only the classification head at 3e-4; full fine-tuning then uses 1e-5 for the backbone and 5e-5 for the head, AdamW with weight decay 0.05, linear warm-up followed by cosine decay, gradient clipping at norm 1.0, and mixed precision. Training loss uses cross-entropy with label smoothing 0.10, while validation loss has no smoothing. Early stopping begins at epoch 15 with patience 8 and minimum Macro-F1 improvement 0.001. Each fold keeps the checkpoint with the highest validation Macro F1.

For final prediction, the five best fold models independently produce class probabilities for the untouched 1,506-image test set. Their probability vectors are averaged and the maximum-probability class becomes the ensemble prediction. This five-model probability ensemble is the primary ViT result used in the comparison.

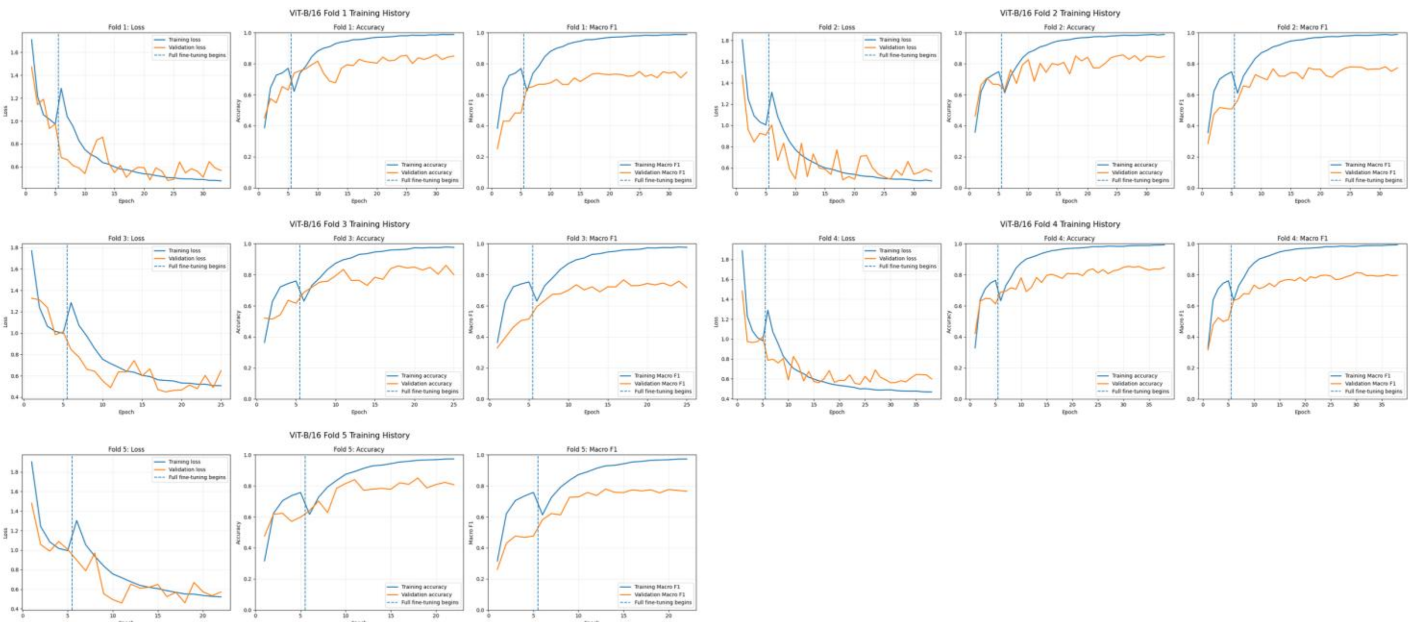

Figure 6. ViT-B/16 per-fold training curves (loss, accuracy, and Macro F1) extracted from the five-fold notebook.

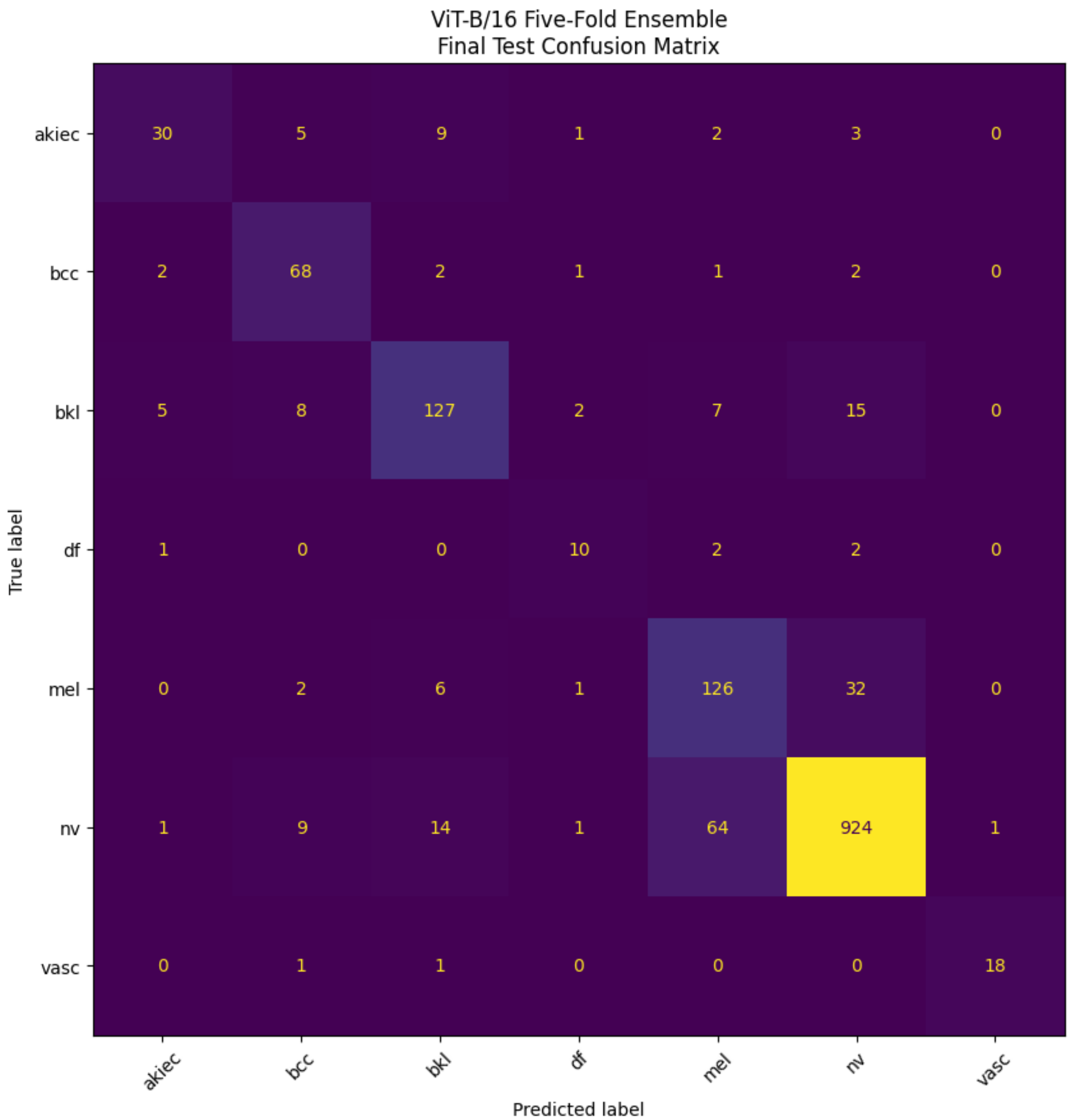

Figure 7. ViT-B/16 five-fold ensemble raw final-test confusion matrix.

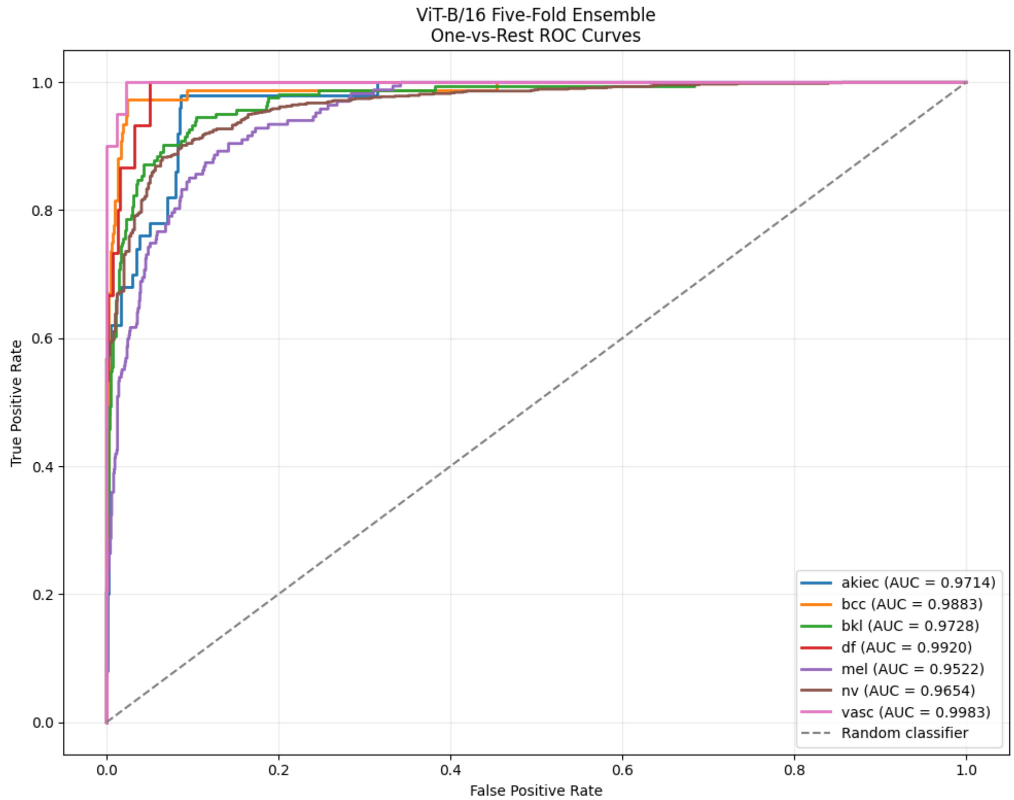

Figure 8. ViT-B/16 five-fold ensemble one-vs-rest ROC curves.

3.6.2 ViT-B/16 evaluation output

Metric	Value
Test protocol	Seed-171 lesion-level final test; five-fold probability ensemble
Test images	1,506
Overall accuracy	0.8652 (86.52%)
Macro precision	0.7772
Macro recall	0.7859
Macro F1	0.7777
Weighted F1	0.8677
Macro ROC-AUC	0.9772

Class	Support	Class-wise accuracy	Precision	Recall	F1-score	OvR AUC
akiec	50	0.6000	0.7692	0.6000	0.6742	0.9714
bcc	76	0.8947	0.7312	0.8947	0.8047	0.9883
bkl	164	0.7744	0.7987	0.7744	0.7864	0.9728
df	15	0.6667	0.6250	0.6667	0.6452	0.9920
mel	167	0.7545	0.6238	0.7545	0.6829	0.9522
nv	1014	0.9112	0.9448	0.9112	0.9277	0.9654
vasc	20	0.9000	0.9474	0.9000	0.9231	0.9983

Class-wise accuracy is defined here as the fraction of correctly classified samples within each true class (TP/support); therefore, it is numerically identical to recall for that class. This definition matches the per-class accuracy implementation used in the GNN notebook and can be read directly from the normalized confusion matrices for the image models.

The ViT notebook main metric object contains Macro ROC-AUC = NaN, but its later one-vs-rest ROC section produces seven valid class AUC values. The reported Macro ROC-AUC of 0.9772 in this paper is the arithmetic mean of those seven class-wise AUC values, not a value copied from the earlier NaN summary.

3.7 Baseline Model 4: Swin-Tiny

3.7.1 Architecture, five-fold training, and ensemble

The Swin baseline uses timm model swin_tiny_patch4_window7_224.ms_in22k_ft_in1k with a 768-to-7 classifier, 27,524,737 parameters, dropout 0.10, and drop-path rate 0.10. It uses the same seed-171 fixed 1,506-image final test set and lesion-safe five-fold development protocol as ViT. Inputs are 224 x 224. Training augmentation includes bicubic RandomResizedCrop (0.80-1.00, ratio 0.90-1.10), horizontal and vertical flips, 20-degree rotation, ColorJitter (0.20/0.20/0.20/0.05), and ImageNet normalization. Validation/test preprocessing is deterministic bicubic resize to 224 x 224 followed by ImageNet normalization.

Class imbalance is handled with a WeightedRandomSampler using inverse class frequency. Training uses cross-entropy with 0.10 label smoothing, AdamW with separate decay/no-decay parameter groups, head-only warm-up for 5 epochs at 3e-4, then full fine-tuning with backbone learning rate 1e-5 and head learning rate 5e-5. Weight decay is 0.05, gradients are clipped to norm 1.0, mixed precision is enabled, and a step-level linear warm-up followed by cosine decay is used. Early stopping starts at epoch 15 with patience 8 and minimum improvement 0.001. The primary final result is the mean-probability ensemble of the five fold checkpoints.

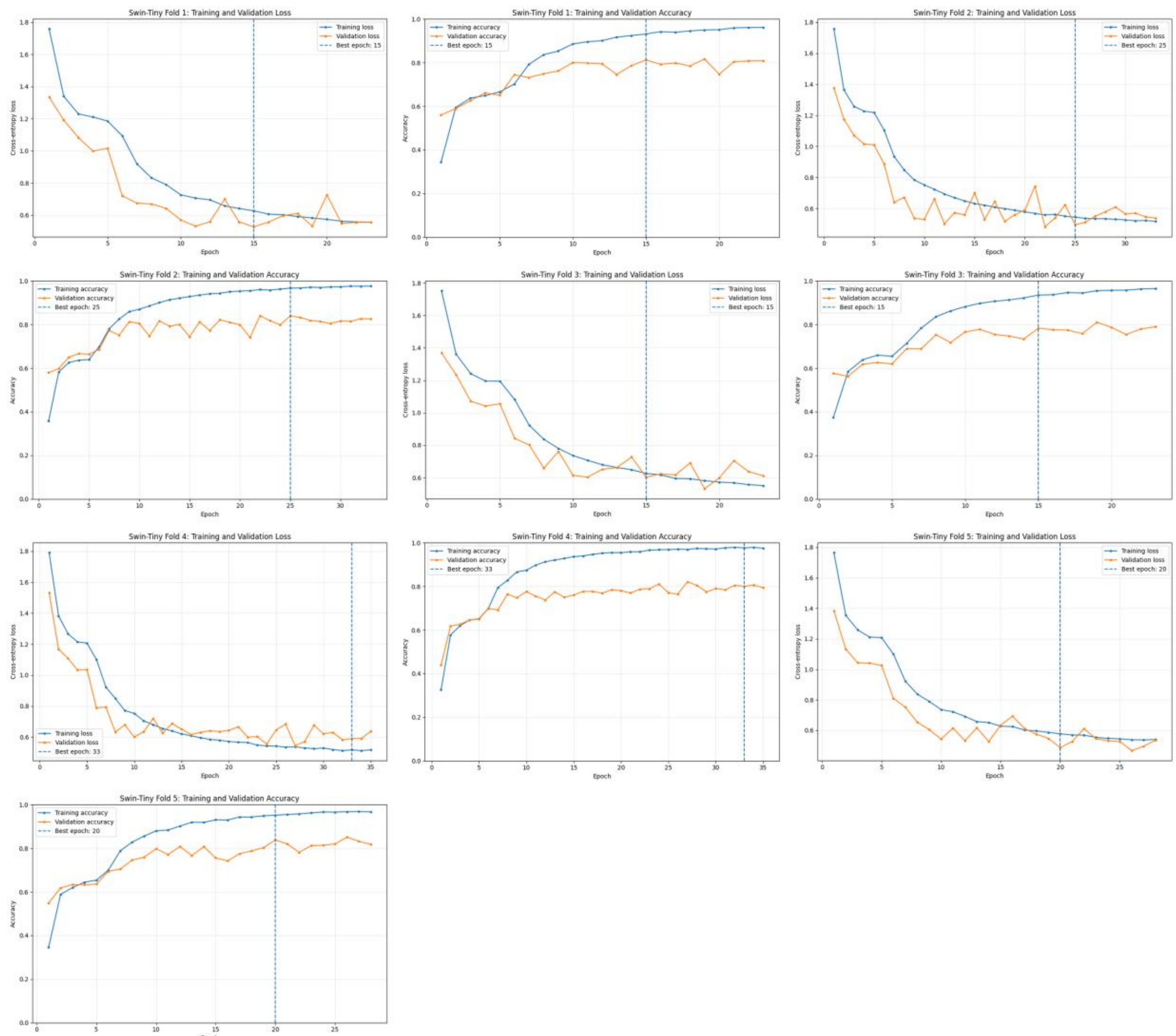

Figure 9. Swin-Tiny five-fold loss and accuracy curves extracted from the notebook (training and validation for each fold).

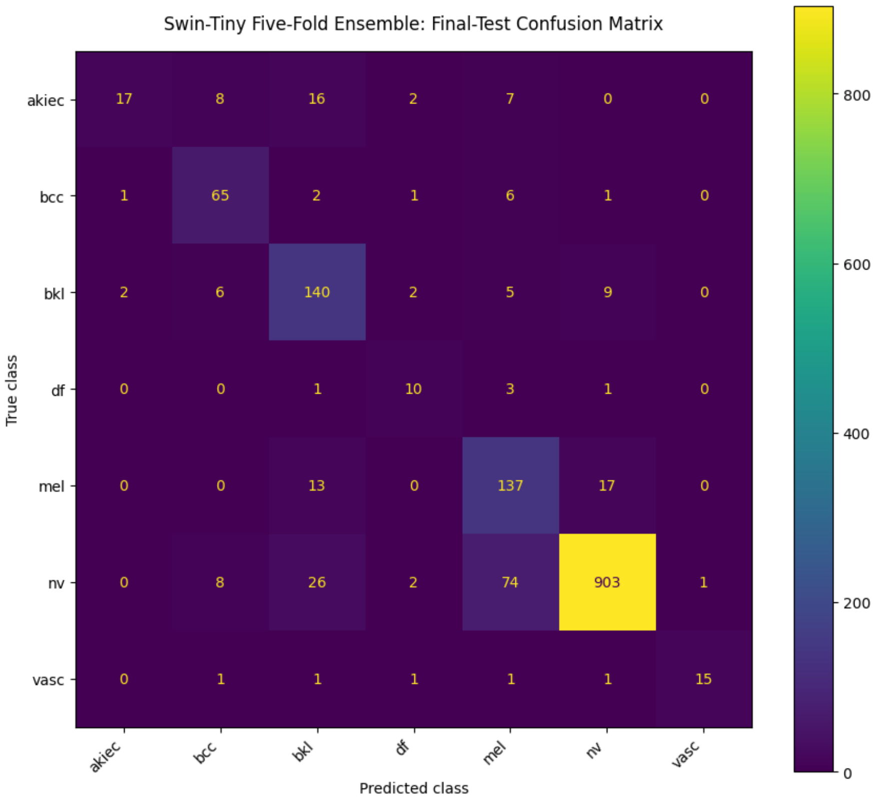

Figure 10. Swin-Tiny five-fold ensemble raw final-test confusion matrix.

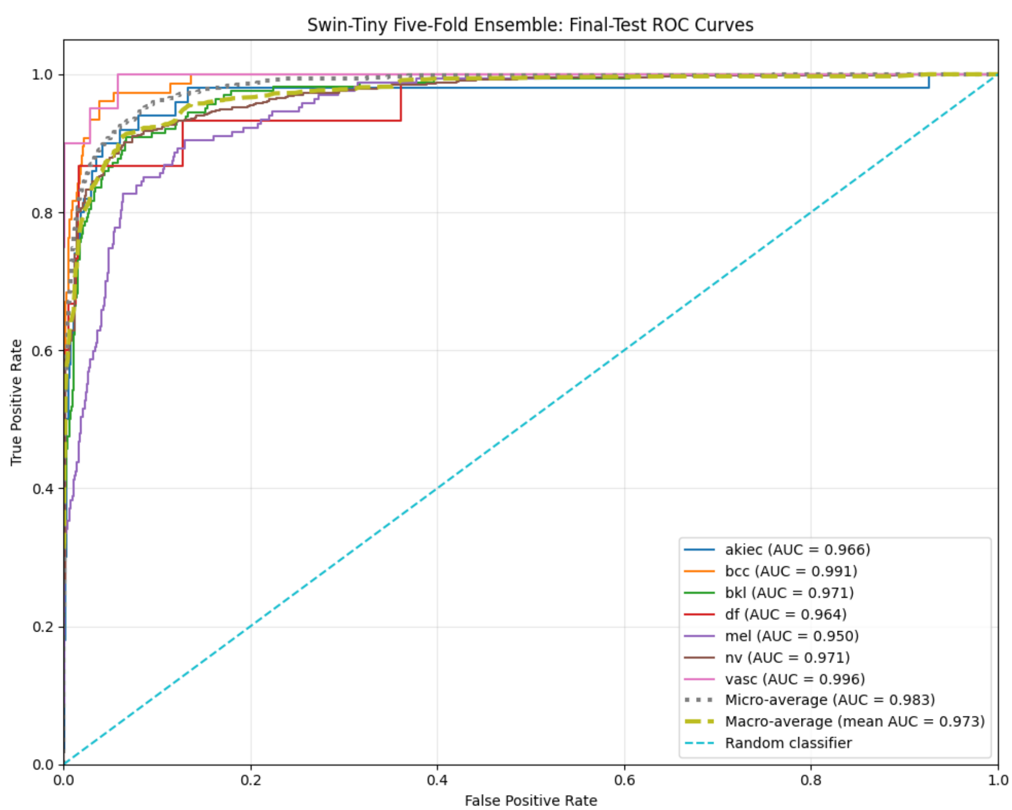

Figure 11. Swin-Tiny five-fold ensemble one-vs-rest ROC curves.

3.7.2 Swin-Tiny evaluation output

Metric	Value
Test protocol	Seed-171 lesion-level final test; five-fold probability ensemble
Test images	1,506
Overall accuracy	0.8546 (85.46%)
Macro precision	0.7632
Macro recall	0.7395
Macro F1	0.7289
Weighted F1	0.8581
Macro ROC-AUC	0.9729

Class	Support	Class-wise accuracy	Precision	Recall	F1-score	OvR AUC
akiec	50	0.3400	0.8500	0.3400	0.4857	0.9663
bcc	76	0.8553	0.7386	0.8553	0.7927	0.9913
bkl	164	0.8537	0.7035	0.8537	0.7713	0.9714
df	15	0.6667	0.5556	0.6667	0.6061	0.9641
mel	167	0.8204	0.5880	0.8204	0.6850	0.9505
nv	1014	0.8905	0.9689	0.8905	0.9281	0.9708
vasc	20	0.7500	0.9375	0.7500	0.8333	0.9956

Class-wise accuracy is defined here as the fraction of correctly classified samples within each true class (TP/support); therefore, it is numerically identical to recall for that class. This definition matches the per-class accuracy implementation used in the GNN notebook and can be read directly from the normalized confusion matrices for the image models.

3.8 Proposed Graph Neural Network Pipeline

3.8.1 Fixed ResNet50 feature encoder

The GNN pipeline first trains a fixed torchvision ResNet50 feature encoder with ImageNet1K_V2 weights. Its custom classifier contains Dropout(0.50), Linear(2048 to 512), ReLU, BatchNorm1d(512), Dropout(0.30), and Linear(512 to 7). Encoder training uses 224 x 224 images, batch size 32, square-root balanced class weights, weighted cross-entropy with 0.05 label smoothing, a 5-epoch classifier warm-up, then partial fine-tuning of layer4 plus the classifier. The encoder is trained once and reused across all GNN experiments to avoid repeatedly recomputing convolutional features.

3.8.2 Image-to-graph construction

Spatial deep features are captured from ResNet50 layer4[-1].bn2. For a 224 x 224 input, the stored feature map is 512 x 7 x 7. Each of the 49 spatial locations becomes one graph node. The 512-dimensional deep feature vector is L2-normalized and concatenated with normalized row and column coordinates, producing a 514-dimensional node feature. The reference graph uses up/down/left/right grid connectivity with self-loops (grid4; 217 directed edges). An eight-neighbor alternative with diagonals and self-loops (grid8; 361 directed edges) is tested during ablation. A cosine-similarity edge-weight option is also evaluated.

3.8.3 Flexible GNN and sequential ablation

The GNN implementation supports GCN, GATv2, GraphSAGE, and SuperGAT while keeping the same cached node features and global mean pooling. Node features first pass through a linear input projection, optional BatchNorm, ReLU and dropout. Each graph-convolution block applies the selected graph operator followed by optional BatchNorm, ELU, a residual connection, and dropout. After global mean pooling, the graph-level classifier uses Dropout, Linear(hidden to 64), ReLU, Dropout, and Linear(64 to 7).

A sequential ablation study changes one factor at a time while selecting the winner exclusively by validation Macro F1 (Macro AUC is used as a tie-breaker). The tested factors are number of layers, hidden dimension, dropout, BatchNorm, learning rate, scheduler, optimizer, graph construction, edge weighting, GNN architecture, class-imbalance loss, and early-stopping patience. The final selected configuration is GATv2 with 2 layers, hidden size 128, 4 attention heads, dropout 0.40, BatchNorm enabled, learning rate 2e-4, cosine scheduler, Adam optimizer with weight decay 5e-4, grid4 graph, no explicit edge weights, weighted cross-entropy with 0.05 label smoothing, and patience 5. The final selected validation Macro F1 is 0.5928.

Ablation factor	Candidates	Selected
GNN layers	2, 3	2
Hidden size	64, 128	128
Dropout	0.30, 0.40, 0.50	0.40
BatchNorm	True, False	True
Learning rate	2e-4, 5e-4	2e-4
Scheduler	Plateau, Cosine	Cosine
Optimizer	AdamW, Adam	Adam
Graph	grid4, grid8	grid4
Edge weights	False, True	False
Variant	GATv2, GCN, GraphSAGE, SuperGAT	GATv2
Imbalance loss	Weighted CE, Focal	Weighted CE
Patience	5, 10	5

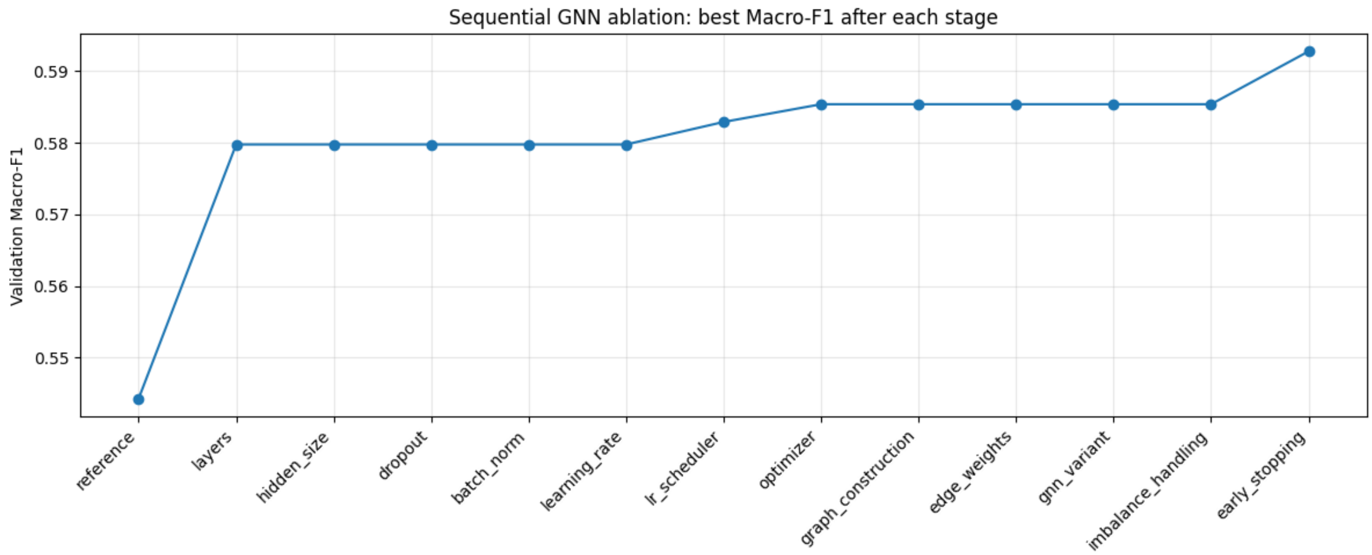

Figure 12. Sequential GNN ablation: best validation Macro F1 after each stage.

3.8.4 Final GNN evaluation and uncertainty

After the final configuration is fixed, a new final GNN is trained on the fixed train/validation protocol and evaluated once on the untouched seed-171 test set. The evaluation reports overall accuracy, balanced accuracy, macro/weighted metrics, per-class accuracy/precision/recall/F1, confusion matrix, one-vs-rest ROC curves, class-wise AUC, training time, and testing time. Because multiple images can belong to one lesion, uncertainty is estimated with 1,000 lesion-level bootstrap resamples rather than image-level resampling. The final GNN 95% bootstrap intervals include accuracy 0.7293-0.7790, Macro F1 0.4846-0.5966, and Macro AUC 0.8440-0.9124.

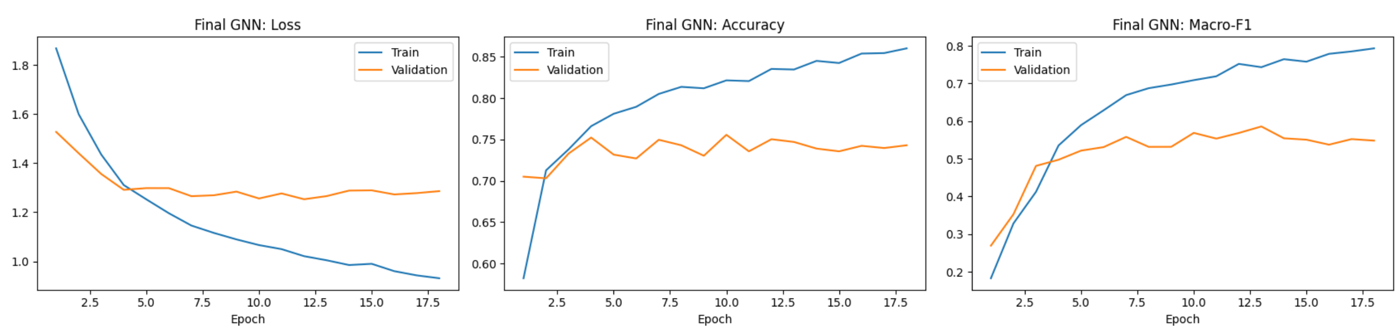

Figure 13. Final GNN training/validation loss, accuracy, and Macro F1 curves.

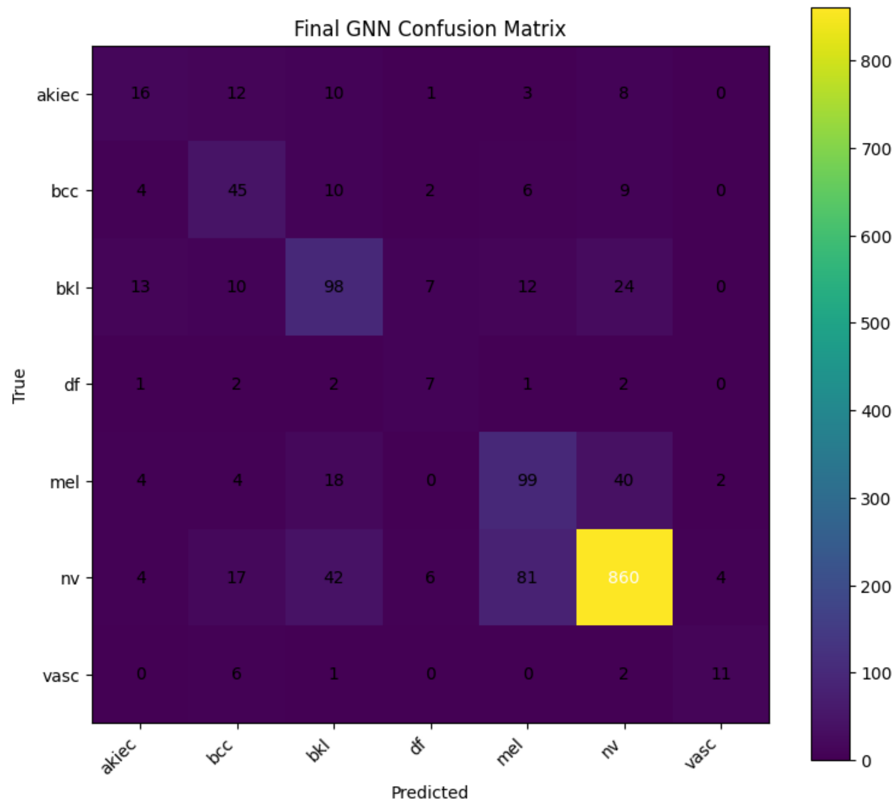

Figure 14. Final GNN raw test confusion matrix.

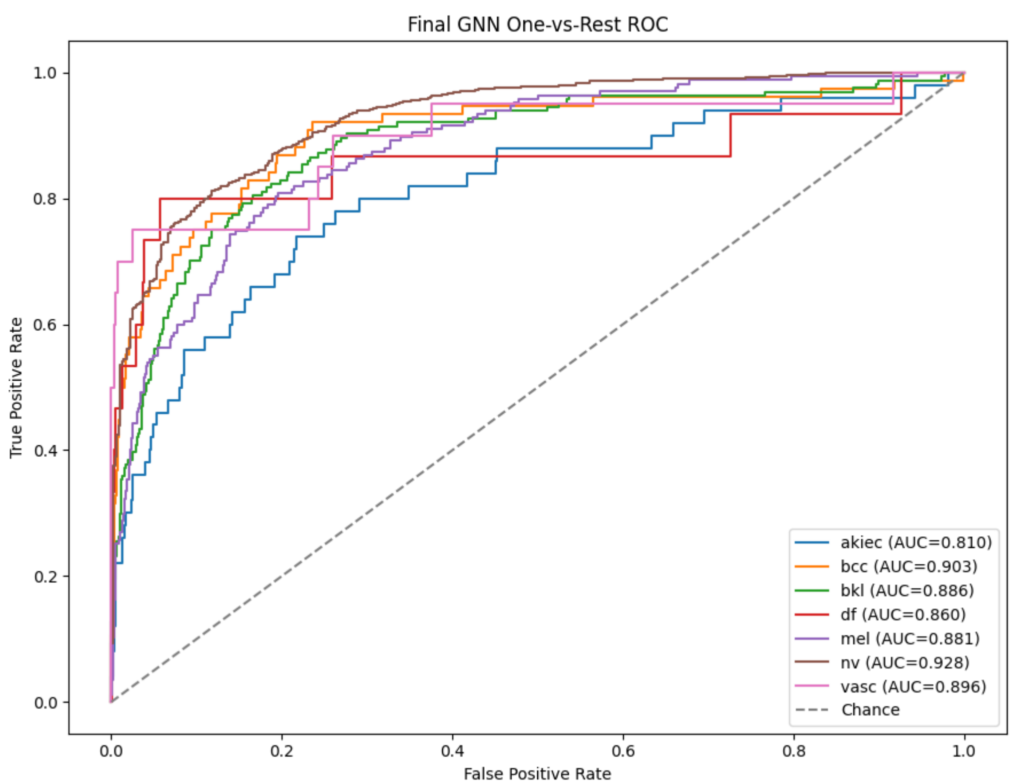

Figure 15. Final GNN one-vs-rest ROC curves.

3.8.5 Final GNN evaluation output

Metric	Value
Test protocol	Seed-171 lesion-level final test; single final GATv2 model
Test images	1,506
Overall accuracy	0.7543 (75.43%)
Balanced accuracy	0.5668
Macro precision	0.5347
Macro recall	0.5668
Macro F1	0.5453
Weighted F1	0.7621
Macro ROC-AUC	0.8806

Class	Support	Class-wise accuracy	Precision	Recall	F1-score	OvR AUC
akiec	50	0.3200	0.3810	0.3200	0.3478	0.8098
bcc	76	0.5921	0.4688	0.5921	0.5233	0.9027
bkl	164	0.5976	0.5414	0.5976	0.5681	0.8860
df	15	0.4667	0.3043	0.4667	0.3684	0.8598
mel	167	0.5928	0.4901	0.5928	0.5366	0.8808
nv	1014	0.8481	0.9101	0.8481	0.8780	0.9284
vasc	20	0.5500	0.6471	0.5500	0.5946	0.8963

Class-wise accuracy is defined here as the fraction of correctly classified samples within each true class (TP/support); therefore, it is numerically identical to recall for that class. This definition matches the per-class accuracy implementation used in the GNN notebook and can be read directly from the normalized confusion matrices for the image models.

3.9 Explainability Procedures

The transformer and graph notebooks contain model-specific explainability outputs. ViT applies attention rollout to visualize spatial attention from a selected fold model. Swin-Tiny applies Grad-CAM, including class-wise correct examples and misclassification analysis. The GNN uses node-masking SHAP and LIME: each explanation treats the 49 spatial graph nodes as interpretable units, keeps or replaces node features with the training-set mean representation, and maps node importance back to a 7 x 7 spatial grid. These figures are interpretation aids rather than medical segmentations or causal explanations.

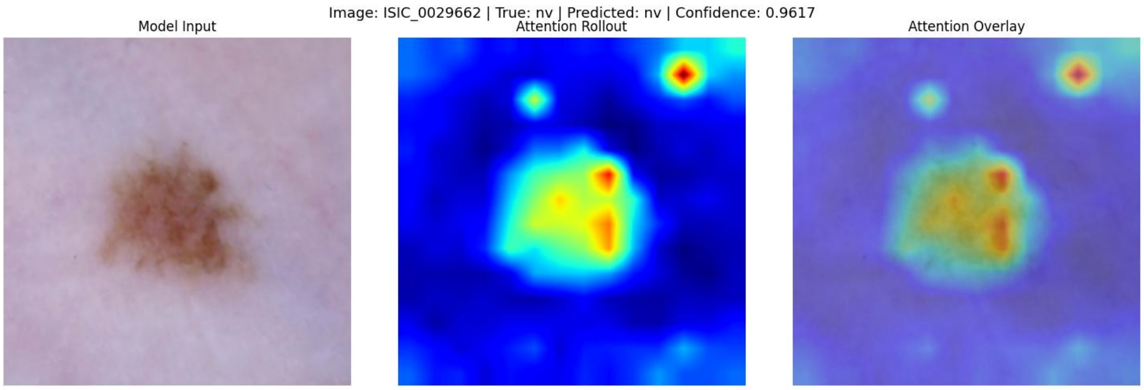

Figure 16. Representative ViT attention-rollout explanation extracted from the notebook.

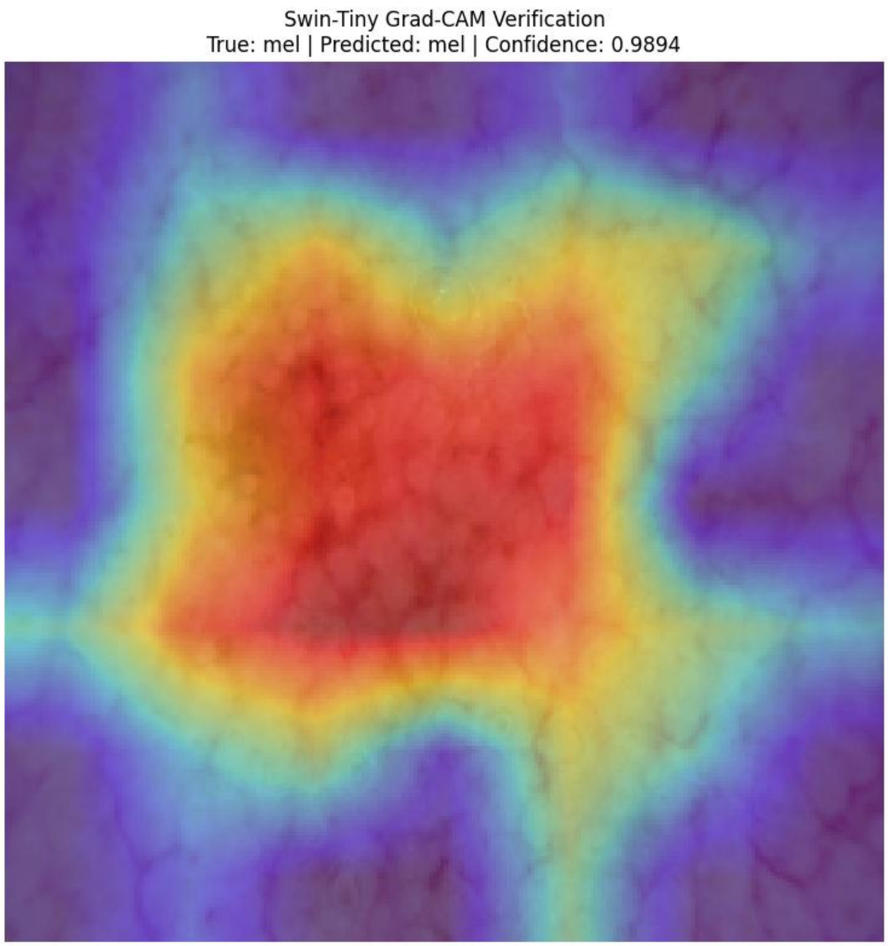

Figure 17. Representative Swin-Tiny Grad-CAM verification output extracted from the notebook.

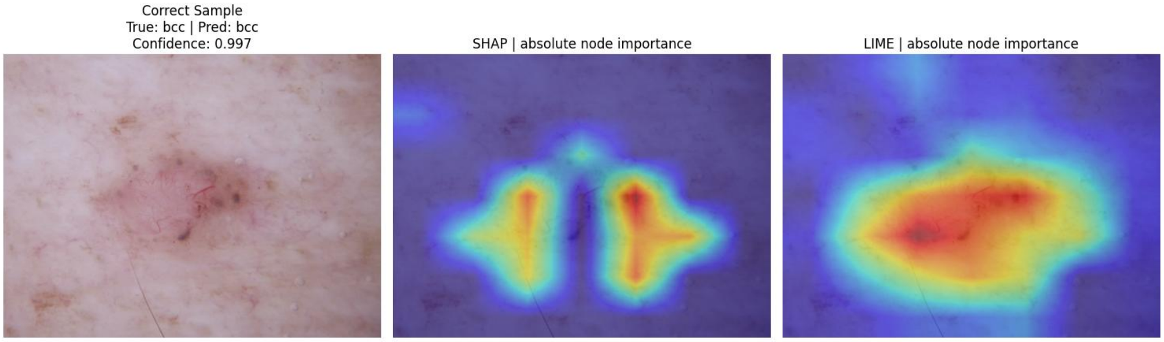

Figure 18. Final GNN SHAP and LIME node-importance maps for a correctly classified example.

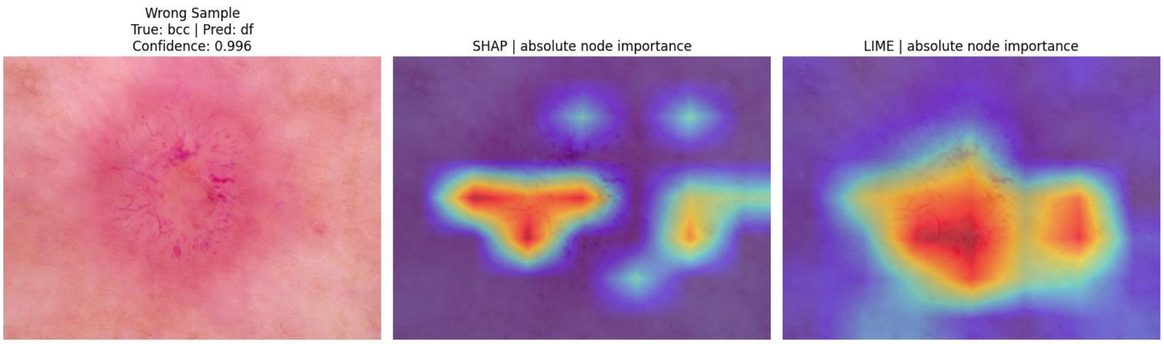

Figure 19. Final GNN SHAP and LIME node-importance maps for a misclassified example.

4. Results and Discussion

4.1 Overall Test Performance

Table 1 summarizes the final metrics from the five model notebooks. The highest raw accuracy and Macro F1 belong to the ViT-B/16 five-model probability ensemble (86.52% accuracy; Macro F1 0.7777). Swin-Tiny is the second strongest model on the same final test (85.46%; Macro F1 0.7289). On the separate seed-42 test set, EfficientNet-B2 reaches 80.52% accuracy and Macro F1 0.6152, exceeding ResNet50 at 78.51% and 0.5887. The Final GNN obtains 75.43% accuracy and Macro F1 0.5453 on the seed-171 test set.

Model	Test n	Accuracy	Macro P	Macro R	Macro F1	Weighted F1	Macro AUC
ResNet50	1,494	0.7851	0.5672	0.6918	0.5887	0.7805	N/A
EfficientNet-B2	1,494	0.8052	0.5956	0.6638	0.6152	0.8050	N/A
ViT-B/16	1,506	0.8652	0.7772	0.7859	0.7777	0.8677	0.9772
Swin-Tiny	1,506	0.8546	0.7632	0.7395	0.7289	0.8581	0.9729
Final GNN	1,506	0.7543	0.5347	0.5668	0.5453	0.7621	0.8806

Table 1 must be interpreted with the split protocol in Section 3.2. ViT, Swin and GNN share the same 1,506-image test set; ResNet50 and EfficientNet-B2 share a different 1,494-image test set.

4.2 Direct Same-Test Comparison: ViT-B/16, Swin-Tiny, and GNN

The strictest comparison is among ViT-B/16, Swin-Tiny and the Final GNN because all three use the identical seed-171 final test set. ViT is best on both overall accuracy and Macro F1. Its Macro F1 exceeds Swin by about 0.0488 and the GNN by about 0.2324. Swin remains close to ViT in overall accuracy but has a lower Macro F1, showing that its class balance is less consistent. The GNN is substantially lower on both global and macro metrics despite using explicit spatial graph reasoning.

The same-test class-wise F1 results show that ViT is strongest on akiec, bcc, bkl, df and vasc, while Swin is marginally stronger on mel and nv. The GNN is below both transformer ensembles in every class on this test. The largest practical challenge is minority-class performance: df has only 15 samples and vasc only 20 in the seed-171 test set, so small changes in correct predictions cause large swings in class-wise metrics.

Class	ViT F1	Swin F1	GNN F1
akiec	0.6742	0.4857	0.3478
bcc	0.8047	0.7927	0.5233
bkl	0.7864	0.7713	0.5681
df	0.6452	0.6061	0.3684
mel	0.6829	0.6850	0.5366
nv	0.9277	0.9281	0.8780
vasc	0.9231	0.8333	0.5946

4.3 ResNet50 versus EfficientNet-B2 on the Seed-42 Holdout

Because ResNet50 and EfficientNet-B2 use the same split, their difference is directly interpretable. EfficientNet-B2 improves accuracy from 0.7851 to 0.8052 and Macro F1 from 0.5887 to 0.6152 while using only about 7.71 million parameters compared with 23.52 million for ResNet50. The gain is visible in bcc, bkl, mel, nv and vasc F1, although EfficientNet-B2 is weaker on the very small df class (0.3200 versus 0.4444). This illustrates why the macro and class-wise tables are needed alongside overall accuracy.

4.4 Training Dynamics and Generalization

The ResNet50 and EfficientNet-B2 curves show the intended two-stage transfer-learning behavior: head-only training for five epochs followed by full fine-tuning. In both cases training loss continues to decrease after the fine-tuning transition while validation loss flattens, and the training accuracy/F1 increasingly exceed validation performance. This indicates a growing generalization gap. EfficientNet-B2 nevertheless stabilizes at a higher validation level than ResNet50 and achieves the stronger seed-42 test result.

ViT and Swin use early stopping inside each fold rather than forcing all models to epoch 50. ViT folds stop after 22-38 epochs, with best epochs ranging from 13 to 31; its five-fold validation Macro F1 mean is 0.7784 +/- 0.0238. Swin best epochs range from 15 to 33, with fold-mean validation Macro F1 0.7394 +/- 0.0289 and pooled OOF Macro F1 0.7404. The probability ensembles provide a small but useful improvement over a single selected fold model, especially for overall robustness.

4.5 GNN Ablation: What Improved and What Did Not

The GNN ablation shows that additional graph complexity was not automatically beneficial. Reducing the GATv2 depth from three layers to two produces the largest early improvement in validation Macro F1 (from about 0.5443 to 0.5798), supporting a shallower graph model for the 7 x 7 node grid. Hidden size 128, dropout 0.40 and BatchNorm are retained. Cosine scheduling increases the selected validation Macro F1 to 0.5829, and Adam raises it to 0.5854. Grid8 has slightly higher validation accuracy than grid4 but lower Macro F1, so grid4 is correctly retained under the pre-defined Macro-F1 selection rule. Explicit edge similarity weights, GraphSAGE, SuperGAT and focal loss do not improve the selected metric. Reducing early-stopping patience from 10 to 5 produces the final validation Macro F1 of 0.5928.

The final GNN improves over the initial GATv2 on the test set: accuracy rises from 0.7337 to 0.7543, Macro F1 from 0.5358 to 0.5453, and Macro AUC from 0.8661 to 0.8806. However, balanced accuracy falls from 0.6023 to 0.5668. Class-wise results explain this mixed outcome: several classes improve modestly, but df F1 decreases from 0.4390 to 0.3684. Therefore, the final configuration improves overall performance without uniformly improving all classes.

Several implementation choices can explain why the graph model does not surpass the transformer baselines. First, the 224 x 224 image is compressed to a 7 x 7 grid of only 49 nodes, which can discard fine lesion detail. Second, ResNet spatial features are cached and fixed during GNN ablation, so the CNN encoder cannot adapt end-to-end to the graph objective. Third, grid4 connectivity is purely spatial and may miss semantically related but non-adjacent regions. Fourth, global mean pooling can dilute highly informative local nodes. Finally, ViT and Swin report five-model ensembles whereas the Final GNN is a single model.

4.6 ROC and AUC Interpretation

ViT, Swin and GNN provide usable one-vs-rest ROC curves. The mean of ViT class-wise AUCs is approximately 0.9772, Swin reports Macro AUC 0.9729, and the GNN reports 0.8806. The ROC figures show high separability for the transformer ensembles, including strong minority-class AUCs, even when the corresponding F1 scores are lower. This is expected because AUC evaluates ranking across thresholds whereas F1 is based on a single final decision threshold.

No valid ResNet50 or EfficientNet-B2 ROC curve is included because the executed notebooks did not produce one. Both final evaluations explicitly caught a multiclass ROC-AUC probability-format error and stored the aggregate AUC as None. Treating those missing outputs as real AUC values would be incorrect.

4.7 Explainability Findings

The explainability outputs demonstrate complementary views of model behavior. ViT attention rollout shows where token-level attention concentrates across the image. Swin Grad-CAM provides class-specific activation maps from the final spatial feature representation. GNN SHAP and LIME identify graph nodes whose retained or masked deep features most influence a prediction. In the correctly classified GNN example, the high-importance regions are concentrated over lesion-related spatial areas; in the wrong example, important regions are more diffuse and can reflect ambiguous visual evidence. These visualizations increase transparency, but none of the notebooks validates them against dermatologist-drawn lesion masks, so they should not be interpreted as clinically verified localization.

4.8 Study Limitations

The principal experimental limitation is that the five reported models are not all evaluated under one identical training and test protocol. ResNet50 and EfficientNet-B2 use a seed-42 holdout, whereas ViT, Swin and GNN use a seed-171 final test. In addition, ViT and Swin are five-model ensembles while the Final GNN is a single model. The GNN does not use five-fold cross-validation, relying instead on a fixed validation split and lesion-level bootstrap uncertainty. Rare classes have small test support, making their per-class metrics high-variance. Finally, ROC output is missing for ResNet50 and EfficientNet-B2 because of the notebook evaluation error, and XAI outputs are not quantitatively validated against expert annotations.

5. Conclusion

This project evaluated four deep-learning baselines and a graph neural network pipeline for seven-class HAM10000 skin-lesion classification while explicitly controlling lesion-level data leakage. The baseline experiments covered two convolutional architectures (ResNet50 and EfficientNet-B2) and two transformer architectures (ViT-B/16 and Swin-Tiny). The proposed GNN converted cached ResNet50 spatial features into a 49-node image graph and used a sequential ablation study to select a two-layer GATv2 configuration.

Among the models sharing the same seed-171 final test set, the ViT-B/16 five-fold probability ensemble achieved the strongest result with 86.52% accuracy and Macro F1 0.7777, followed by Swin-Tiny with 85.46% accuracy and Macro F1 0.7289. On the separate seed-42 holdout, EfficientNet-B2 outperformed ResNet50 while using considerably fewer parameters. The Final GNN reached 75.43% accuracy, Macro F1 0.5453, and Macro AUC 0.8806. Although it improved over the initial GATv2 configuration, it did not outperform the transformer baselines.

The main value of the GNN experiment is therefore not a claim of superior predictive performance, but a controlled investigation of spatial graph reasoning, architecture choices, class-imbalance handling, uncertainty, and node-level explainability. Future work should test an end-to-end trainable CNN-GNN pipeline, richer graph construction such as semantically adaptive regions, attention-based graph pooling, stronger strategies for rare classes, repeated or cross-validated GNN evaluation, and external clinical validation. In particular, the literature-review motivation for Bangladesh-relevant clinical data remains open because the current experiments are conducted on HAM10000 rather than a locally collected clinical dataset.

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